THE DIAGONAL MORPHISM K -> RHOM(L, K TENSOR L) IN THE DERIVED CATEGORY

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- For K-flat K and any complex L there is a canonical map K -> Hom(L, Tot(K tensor L)). invoked at proof line 18851. Supplies the first arrow of the composite defining the morphism.

Source reference. Original lemma 0BYN: sources/stacks-more-algebra/pinned-original-source.tex, source lines 18036–18036. Cited at proof line 18840 to note the special-case route with M = A; the direct proof given afterwards is inside the counted span.

Source reference. Original lemma 0A62: sources/stacks-more-algebra/pinned-original-source.tex, source lines 18114–18114. Cited at proof line 18851 as the first arrow of the composite defining the morphism; the conclusion is used inside the counted span.

Lemma 0.2. *Let R be a ring. Given complexes K, L in $D(R)$ there is a canonical morphism*

$$K \longrightarrow R\operatorname{Hom}_R(L, K \otimes^{\mathbf{L}}_R L)$$

in $D(R)$ functorial in both K and L.

Proof. This is a special case of Lemma 0BYN but we will also prove it directly. Choose a K-flat complex $K^\bullet$ representing K and any complex $L^\bullet$ representing L. Choose a quasi-isomorphism $\operatorname{Tot}(K^\bullet \otimes_R L^\bullet) \to J^\bullet$ where $J^\bullet$ is K-injective. Then we use the map

$$K^\bullet \to \operatorname{Hom}^\bullet(L^\bullet, \operatorname{Tot}(K^\bullet \otimes_R L^\bullet)) \to \operatorname{Hom}^\bullet(L^\bullet, J^\bullet)$$

where the first map is the map from Lemma 0A62. □

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